

Inverse Problems in Geophysics

Part 11: Probability & Likelihood

2. MGPY+MGIN

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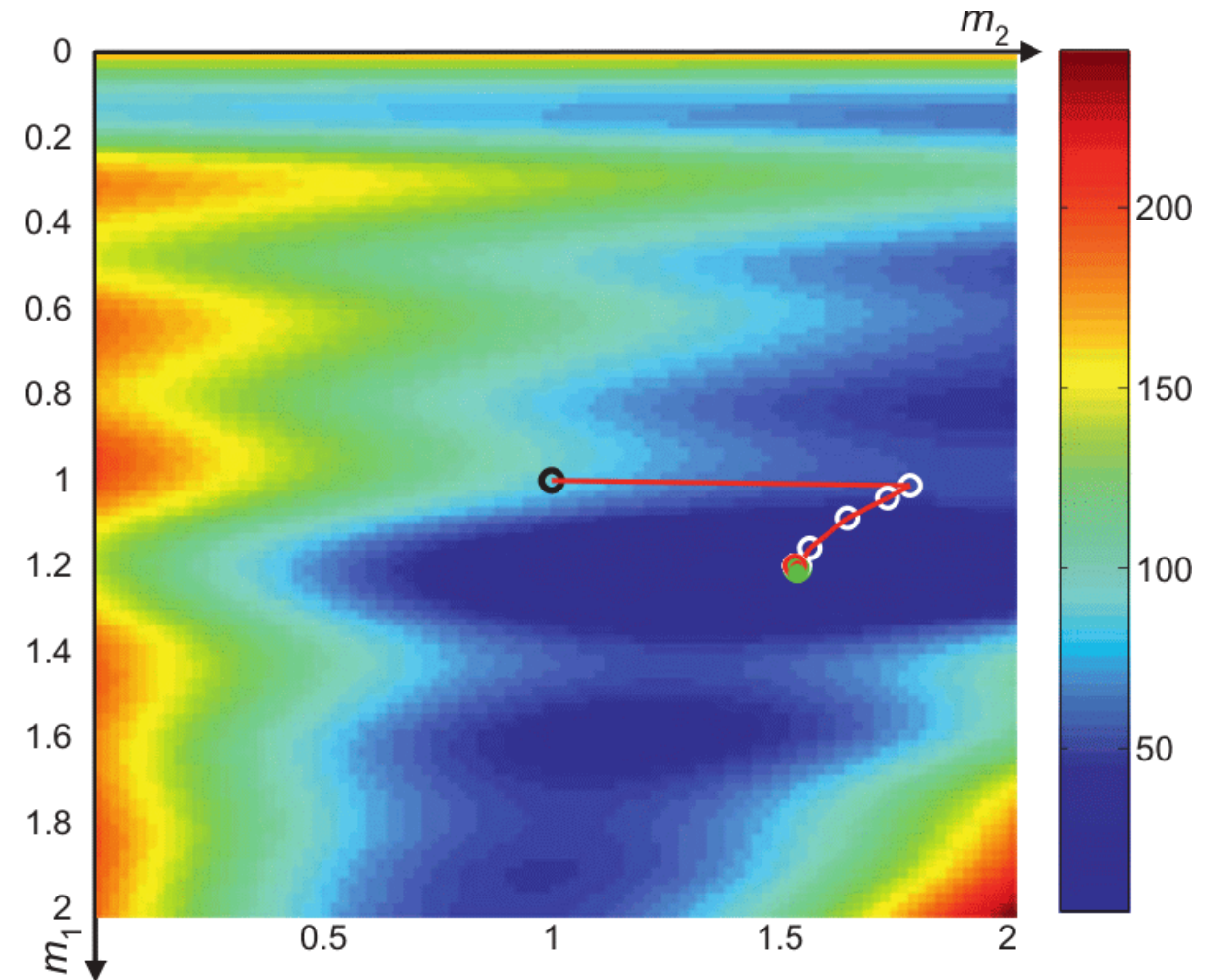
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Recap non-linear inv

undirected (grid) search

directed (gradient) methods

- Steepest descent
- Gauss-Newton method
- depend on starting model
- local damping: Marquardt
- global smoothing: Occam



Gauss-Newton method (Menke, 2015)

Regularization effect

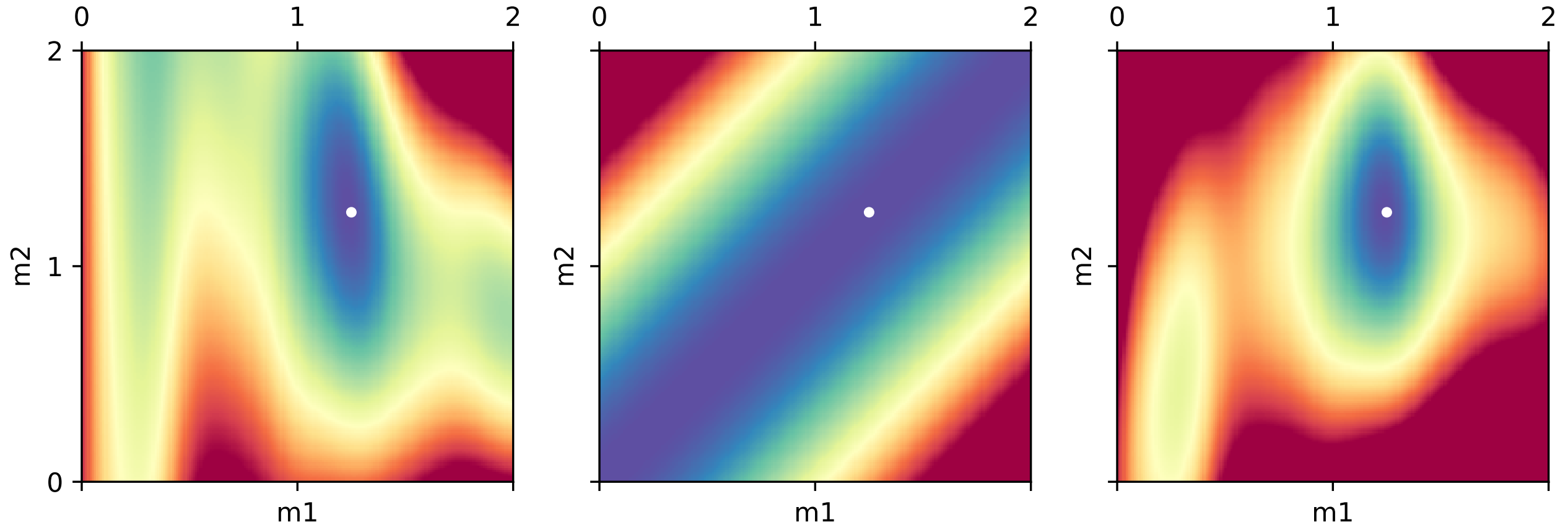
$$\Phi = \Phi_d + \lambda \Phi_m$$

Φ_m : difference from reference model or smoothness

How does this regularization affect convergence?

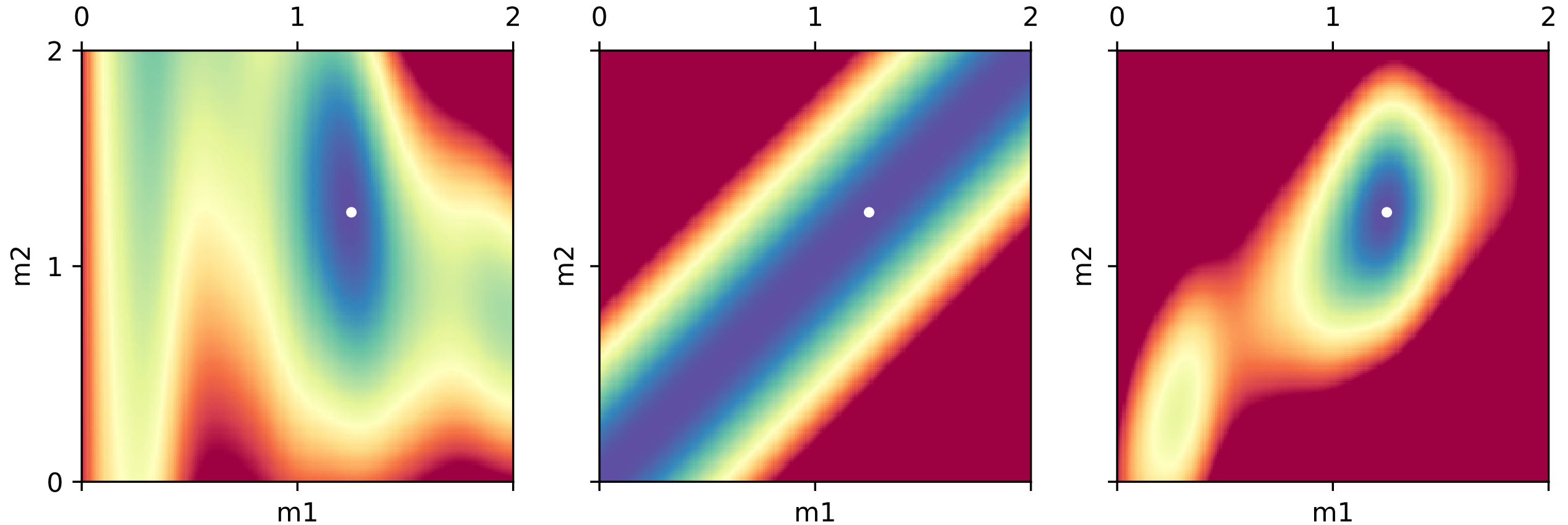
$\Rightarrow$ impact of regularization terms on objective function

Smoothness



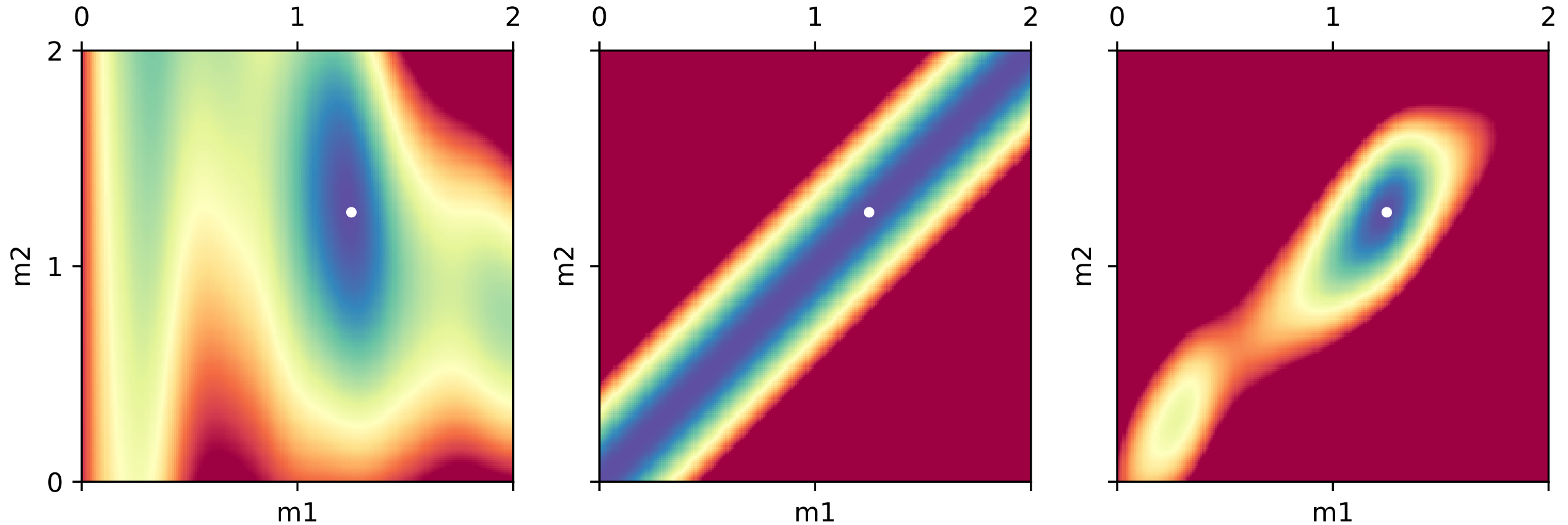
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=30$)

Smoothness



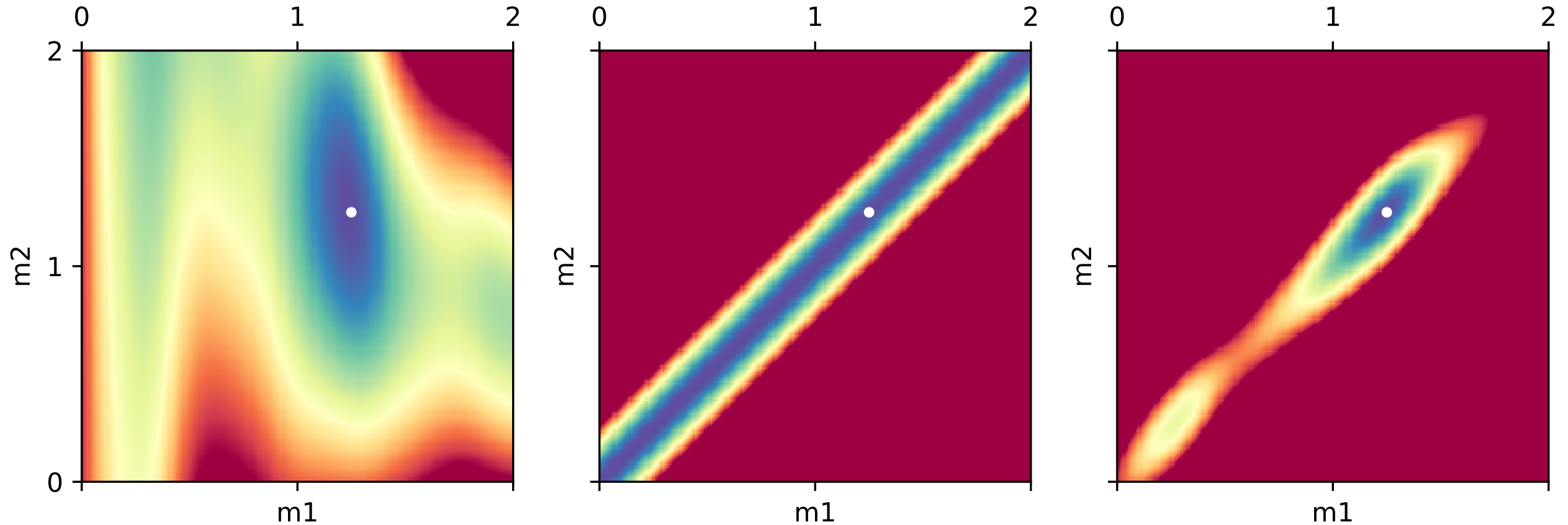
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=100$)

Smoothness



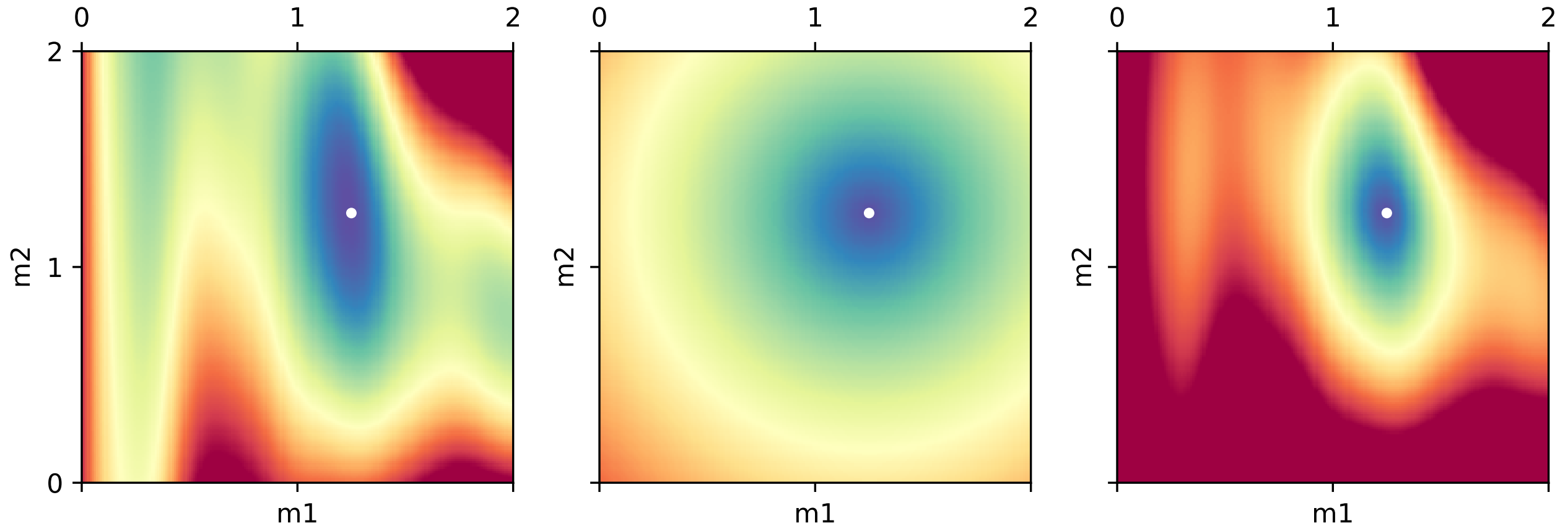
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=300$)

Smoothness



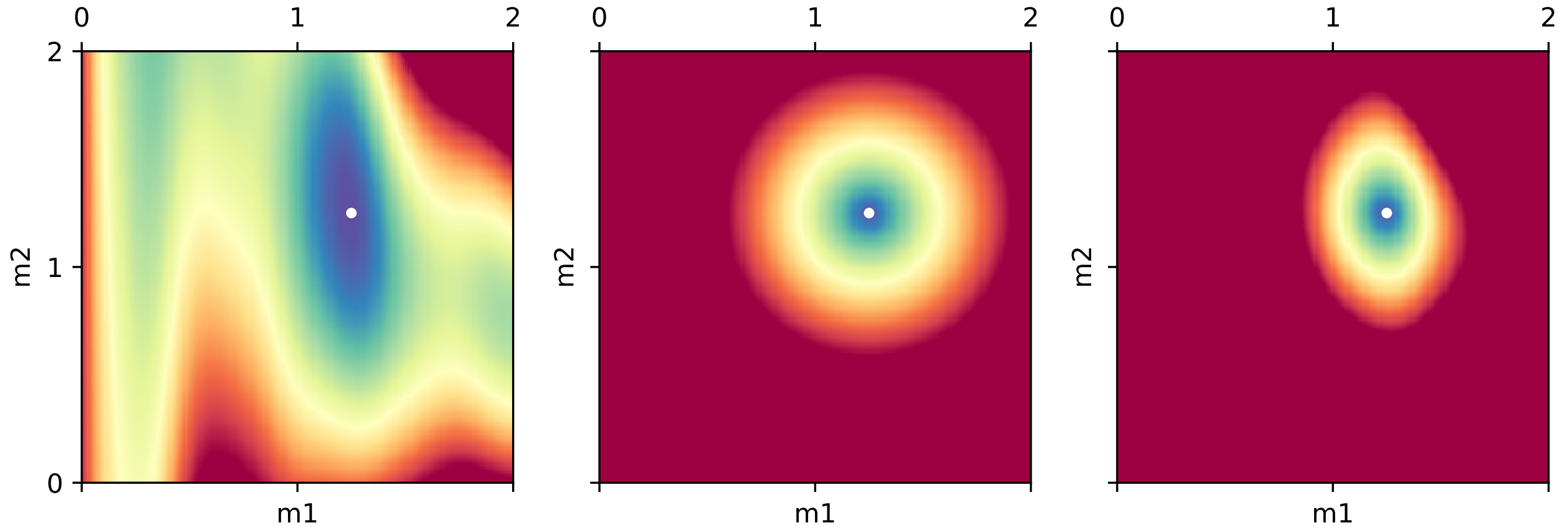
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=1000$)

A-priori information



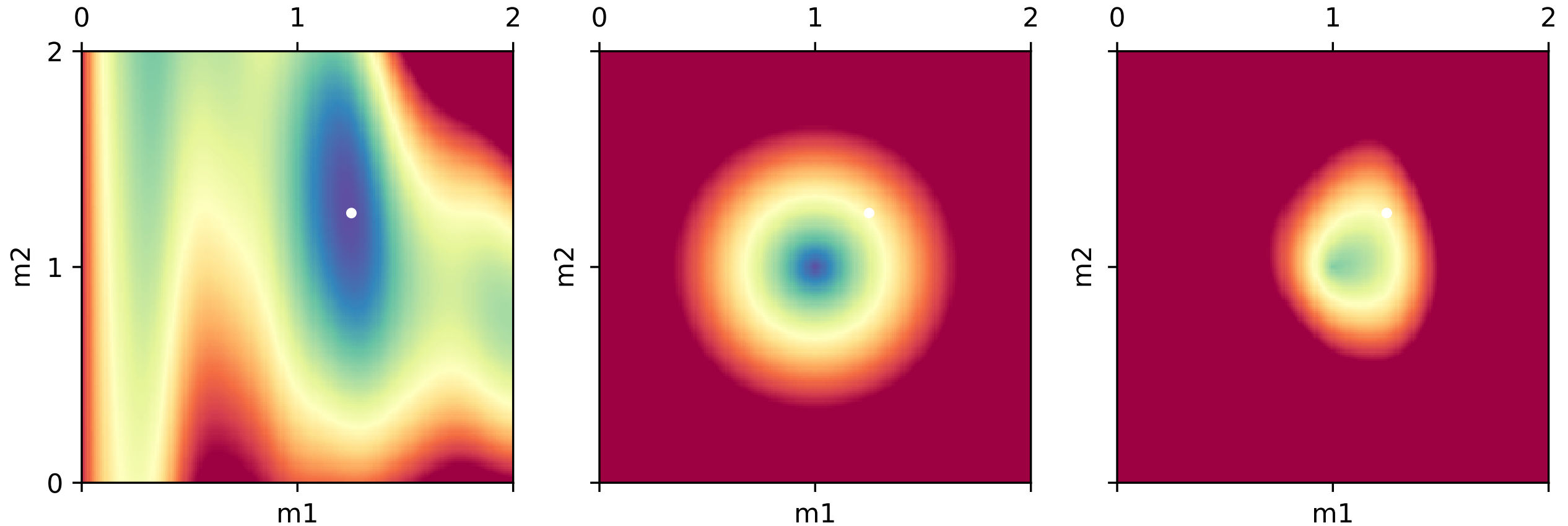
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A-priori information



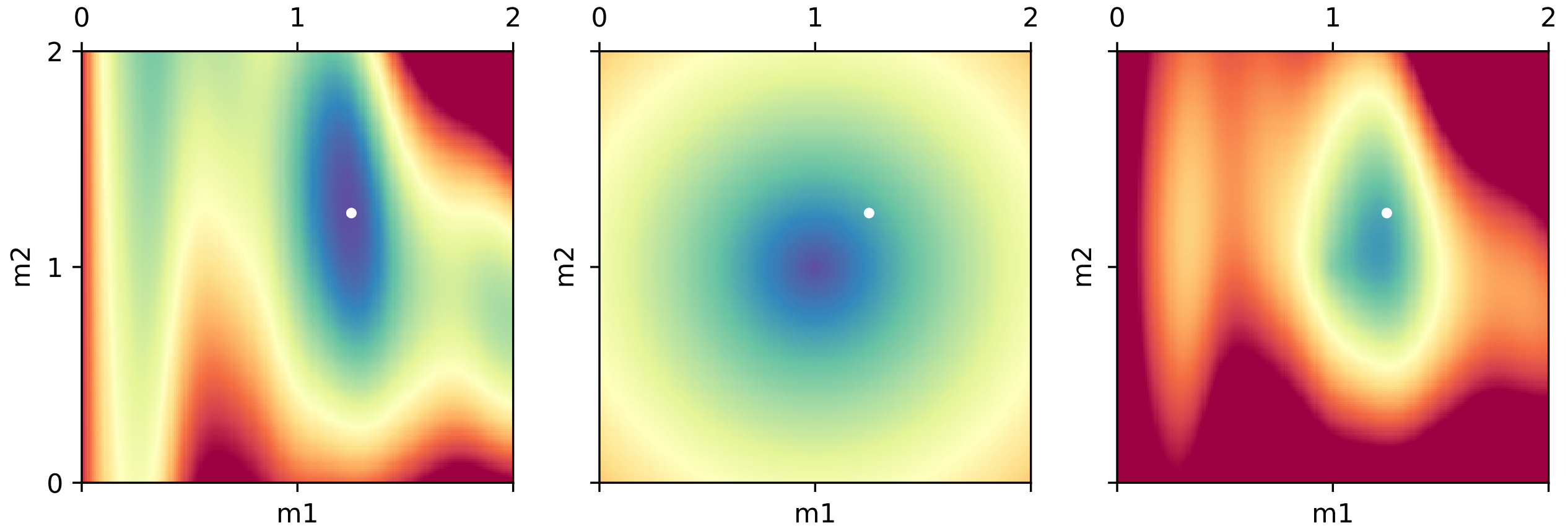
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=100$)

Wrong a-priori information



left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=100$)

Wrong a-priori information



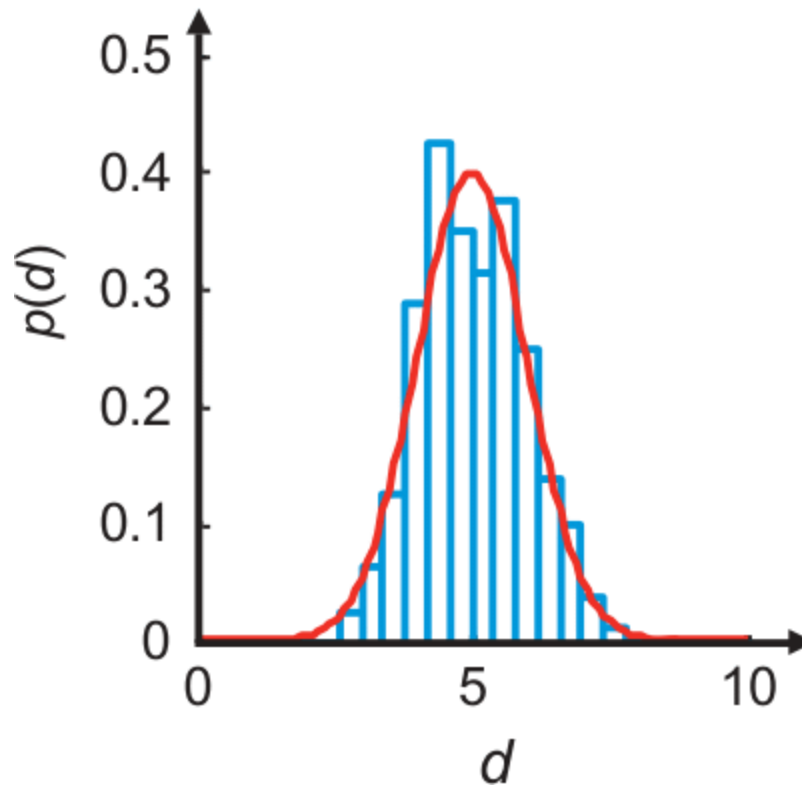
left: Φ_d , center: $\lambda\Phi_m$, right: $\Phi_d + \lambda\Phi_m$ ($\lambda=100$)

Probability and likelihood

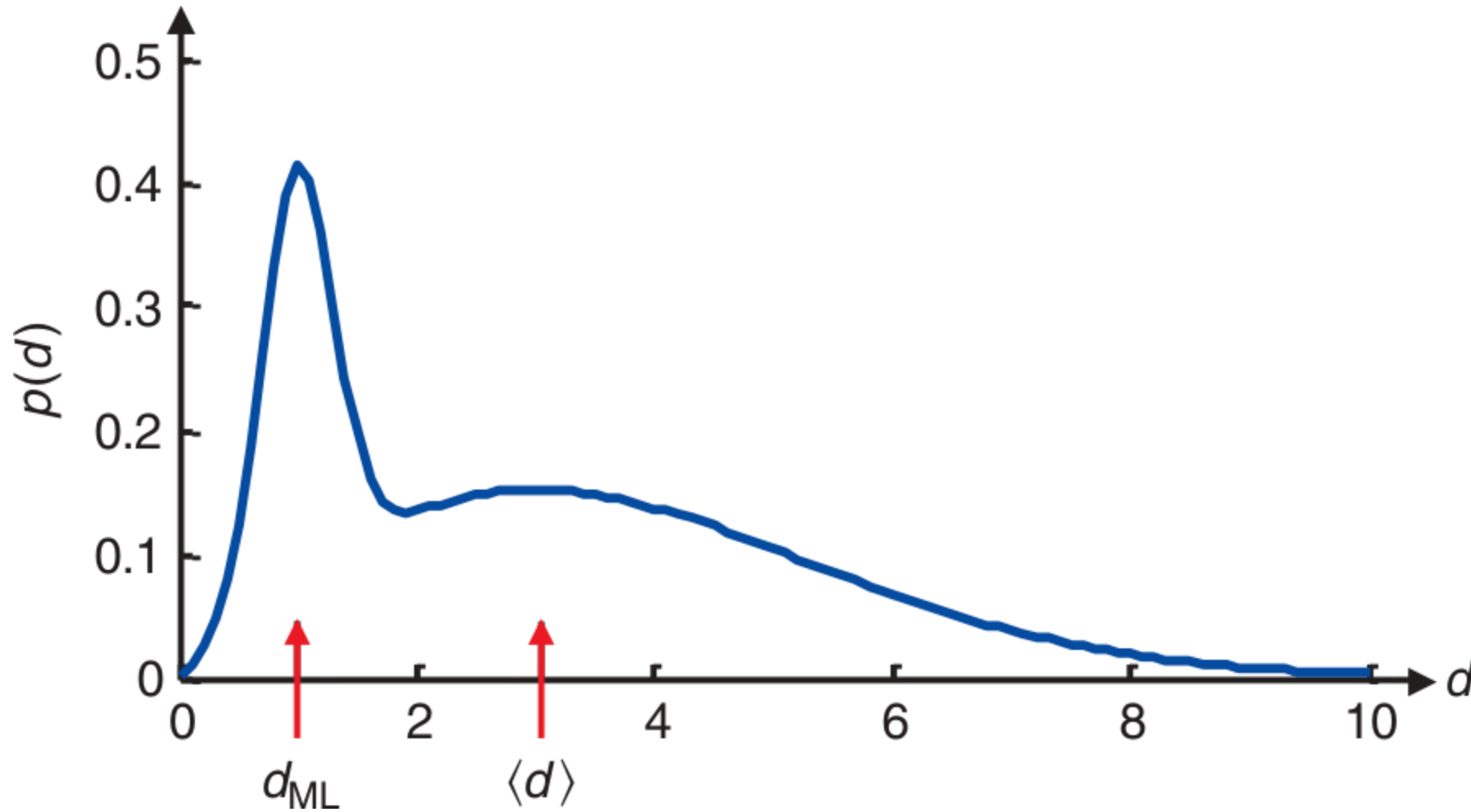
random variables: probability
through many repetitions

maximum $p(d)$ is most likely

expectation: $\langle d \rangle = \int d \cdot p(d) dd$



Probability density function



$$\int p(d) \mathrm{d}d = 1$$

Definitions

! Probability

defines how probable a certain event is expected to occur

Given a known model or system, what are the chances of a specific outcome?

! Likelihood

defines probability of data under the assumption of a model (hypothesis) is true

Given an observed outcome, what is the chance that our model or system parameters are correct?

! Conditional probability

$P(A|B)$ Probability of A under the assumption that B occurs

Variance

$$\sigma^2 = \int (d - \langle d \rangle)^2 p(d) \, dd$$

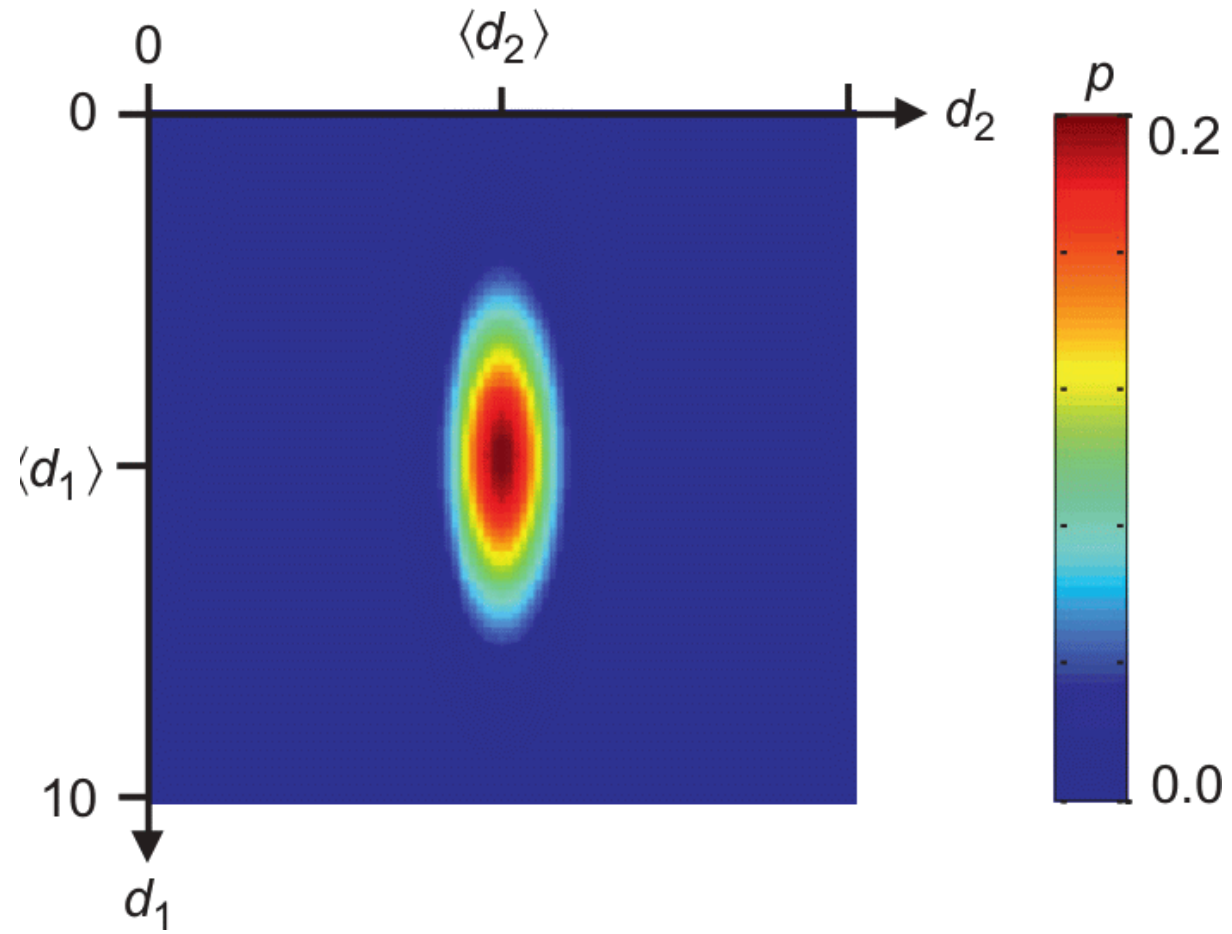
σ is a measure of the width of the distribution

related to standard deviation and mean of sampling

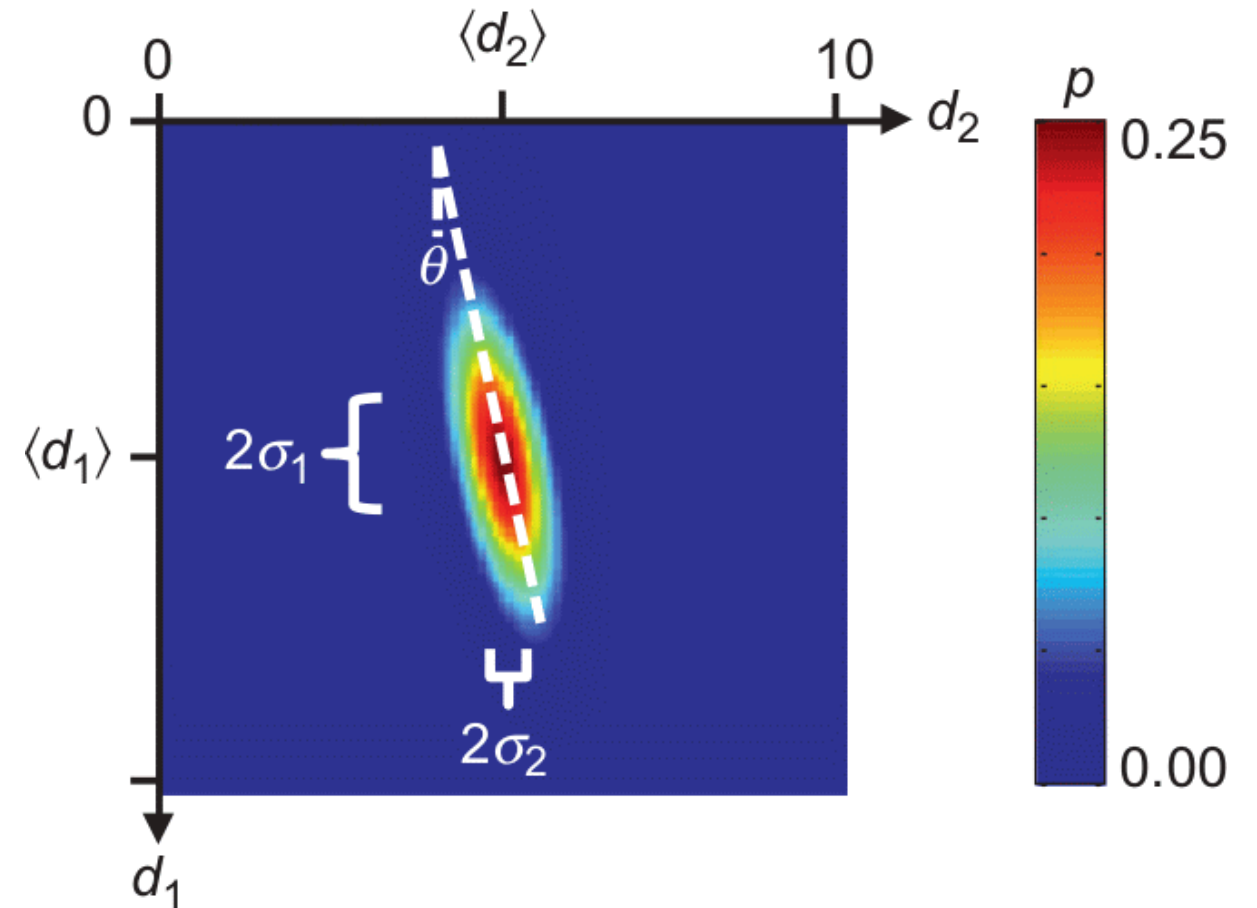
$$\sigma_{est}^2 = \frac{1}{N-1} \sum_{i=1}^N (d_i - \langle d \rangle)^2 \quad \text{with} \quad \langle d \rangle = \frac{1}{N} \sum_{i=1}^N d_i$$

Data correlation

independent: $p(\mathbf{d}) = p(d_1)p(d_2) \dots p(d_N)$



uncorrelated data (Menke, 2012)



correlated data (Menke, 2012)

Covariance

(measure of correlation between data)

$$\text{cov}(d_1, d_2) = \int \int (d_1 - \langle d_1 \rangle)(d_2 - \langle d_2 \rangle) p(d_1, d_2) \, dd_1 \, dd_2$$

$$\langle d_i \rangle = \int \dots \int d_i p(\mathbf{d}) \, dd_1 \dots dd_N$$

Covariance propagation

Linear problem $\mathbf{m} = \mathbf{M}\mathbf{d}$, e.g., $\mathbf{m} = \mathbf{G}^\dagger \mathbf{d}$

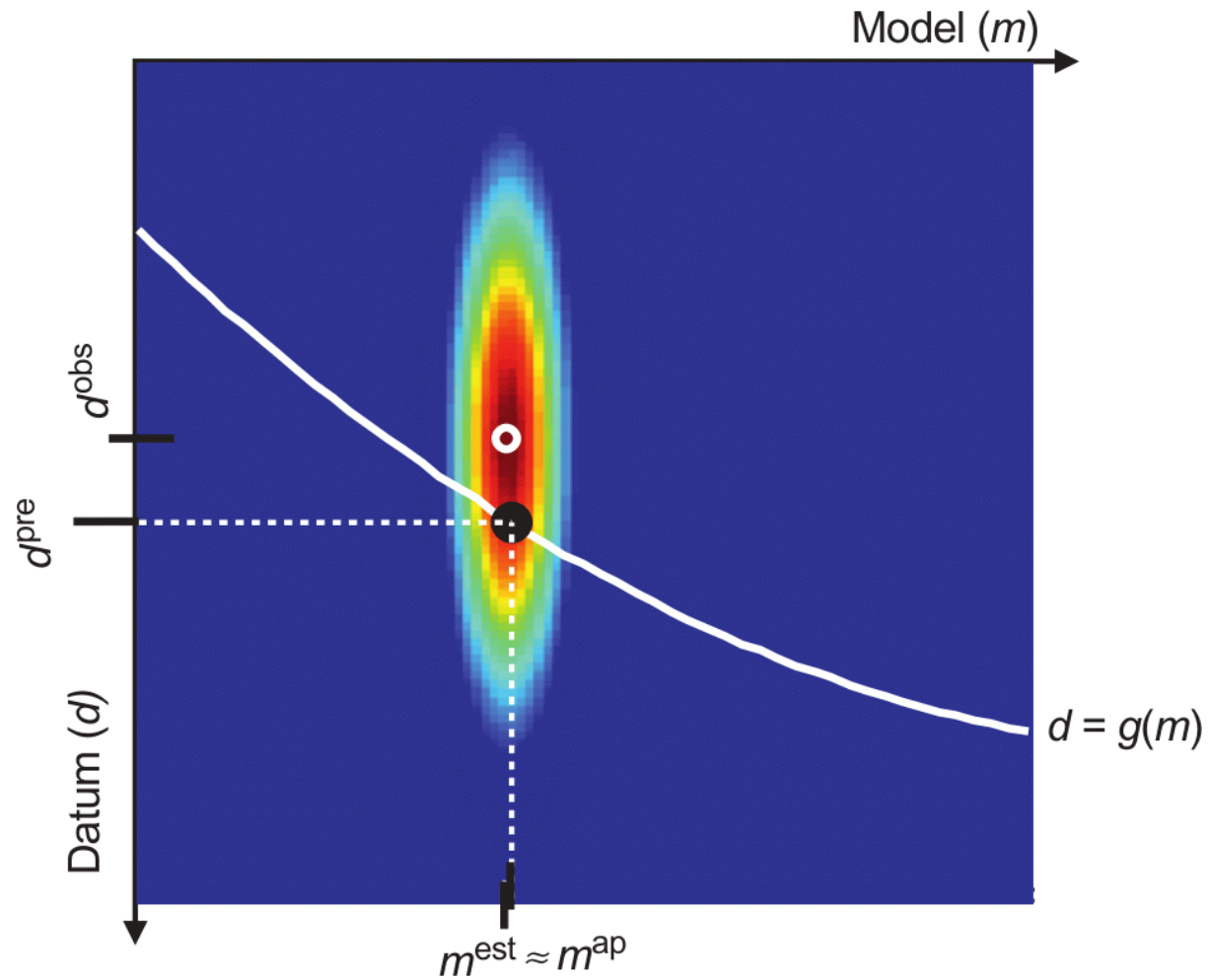
Mean value $\langle \mathbf{m} \rangle = \mathbf{M} \langle \mathbf{d} \rangle + \mathbf{n}$ and covariance

$$\text{cov}(\mathbf{m}) = \mathbf{M} \text{cov}(\mathbf{d}) \mathbf{M}^T$$

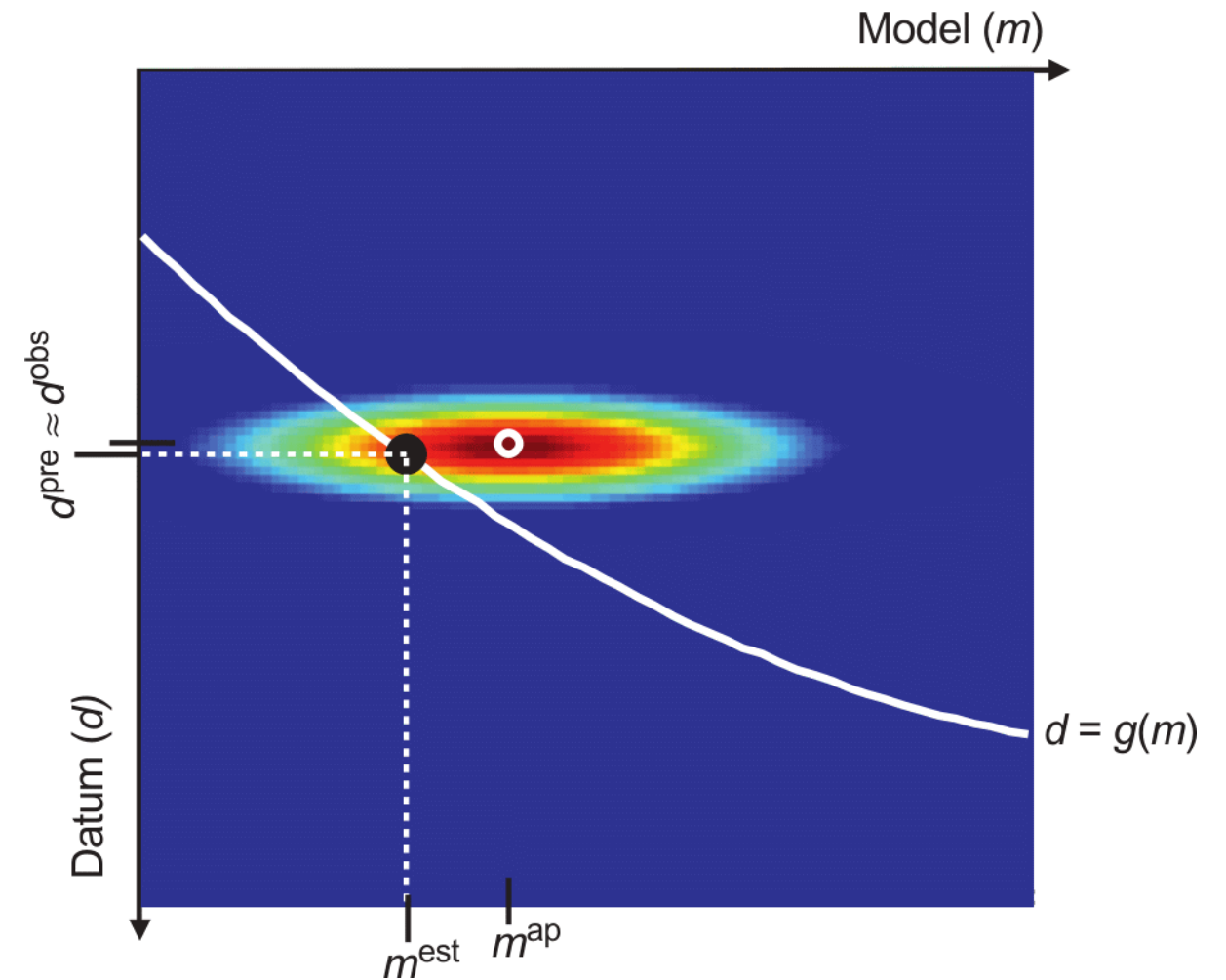
Least-squares: $\mathbf{M} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$, uncorrelated data: $\text{cov}(\mathbf{d}) = \sigma_d^2 \mathbf{I}$

$$\Rightarrow \text{cov}(\mathbf{m}) = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \sigma_d^2 \mathbf{I} ((\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T)^T = \sigma_d^2 (\mathbf{G}^T \mathbf{G})^{-1}$$

A priori knowledge



accurate prior model (Menke, 2012)



accurate data (Menke, 2012)

Bayes' theorem

① Conditional probability

$$p(a|b) = p(a, b)/p(b)$$

$$p(\mathbf{m}|\mathbf{d})p(\mathbf{d}) = p(\mathbf{d}|\mathbf{m})p(\mathbf{m})$$

$$p(\mathbf{m}|\mathbf{d}) = \frac{p(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{p(\mathbf{d})}$$

posterior distribution $\propto$ likelihood x prior distribution

Example: COVID test

- $P(T)$: probability of positive test (T)
- $P(I)$: probability of a person to be ill (1%) (I)
- $P(T|I)$: (conditional) probability of a test recognizing illness (90%)

assume 1000 patients (1%=10 ill, 99%=990 healthy)

false tests (10%) $\Rightarrow$ 1 false negative (9 correct), 99 false positive

$$P(I|T) = \frac{P(T|I)P(I)}{P(T)} = \frac{0.9 \cdot 0.01}{108/1000} = 8.3\%$$

Example: COVID test (2)

- $P(T)$: probability of positive test (T)
- $P(I)$: probability of a person to be ill (1%) (I)
- $P(T|I)$: (conditional) probability of a test recognizing illness (99%)

assume 10000 patients (1%=100 ill, 99%=9900 healthy)

false tests (1%) $\Rightarrow$ 1 false negative (99 correct), 990 false positive

$$P(I|T) = \frac{P(T|I)P(I)}{P(T)} = \frac{0.99 \cdot 0.01}{1089/10000} = 9.1\%$$

FIFA example

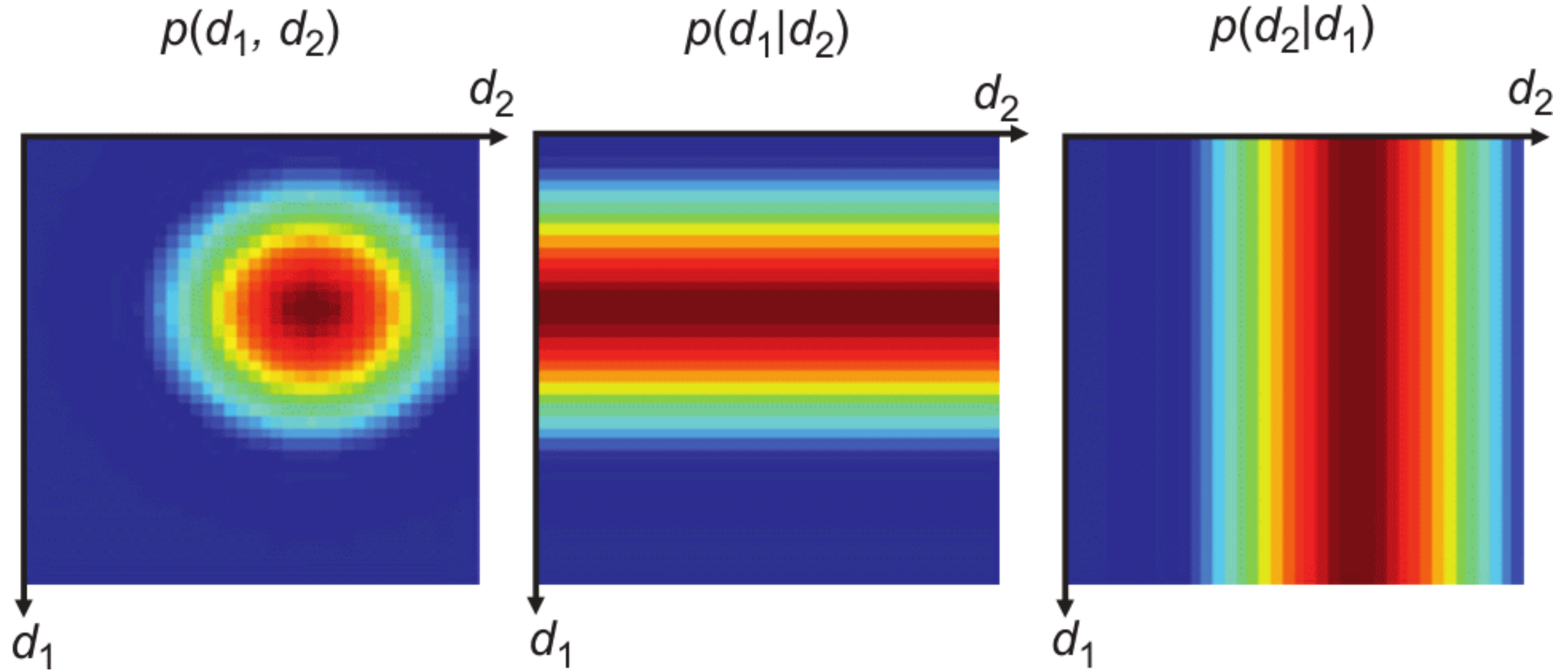
Note

Tony hears football-watching Arthur cheer. How probable is a goal being made?

1. Assumption: In 2% of the time (segments covering a cheer) there is a goal.
2. Assumption: If a goal is made by Arthurs team, he cheers by 90%.
3. Assumption: Reasons for non-goal cheers (98%) have a probability of 1%.

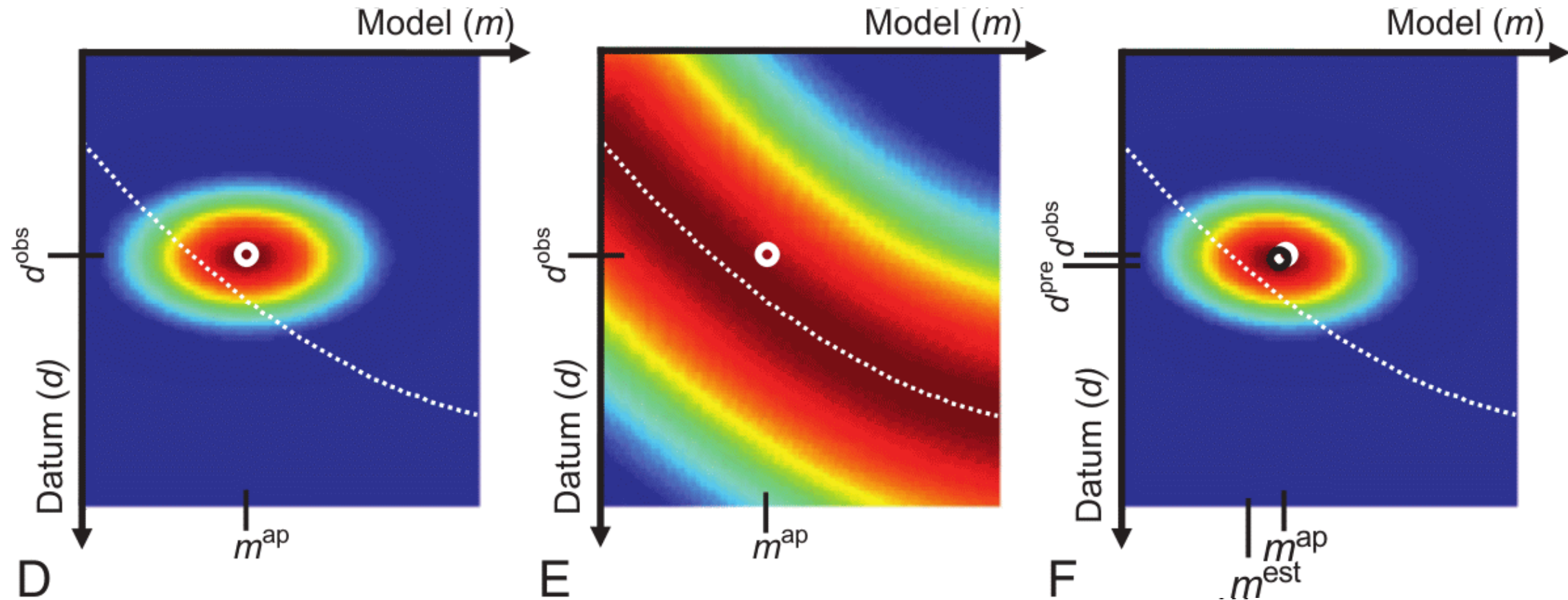
$$P(G|C) = \frac{P(C|G)P(G)}{P(C|G) + P(C| - G)} = \frac{0.9 \cdot 0.02}{0.02 \cdot 0.9 + 0.98 \cdot 0.01} = 64.7\%$$

Bayes theorem simple example



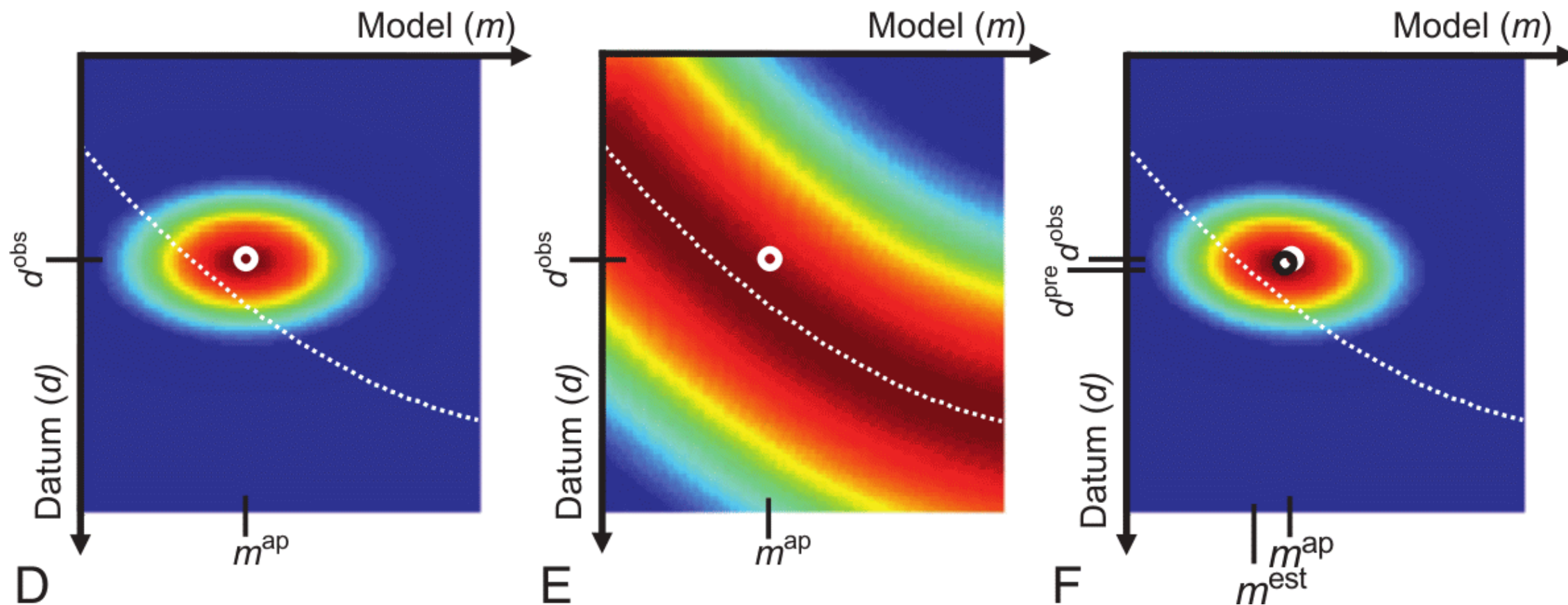
joint probability & conditional probabilities (Menke, 2015)

A priori (Menke, 2012)

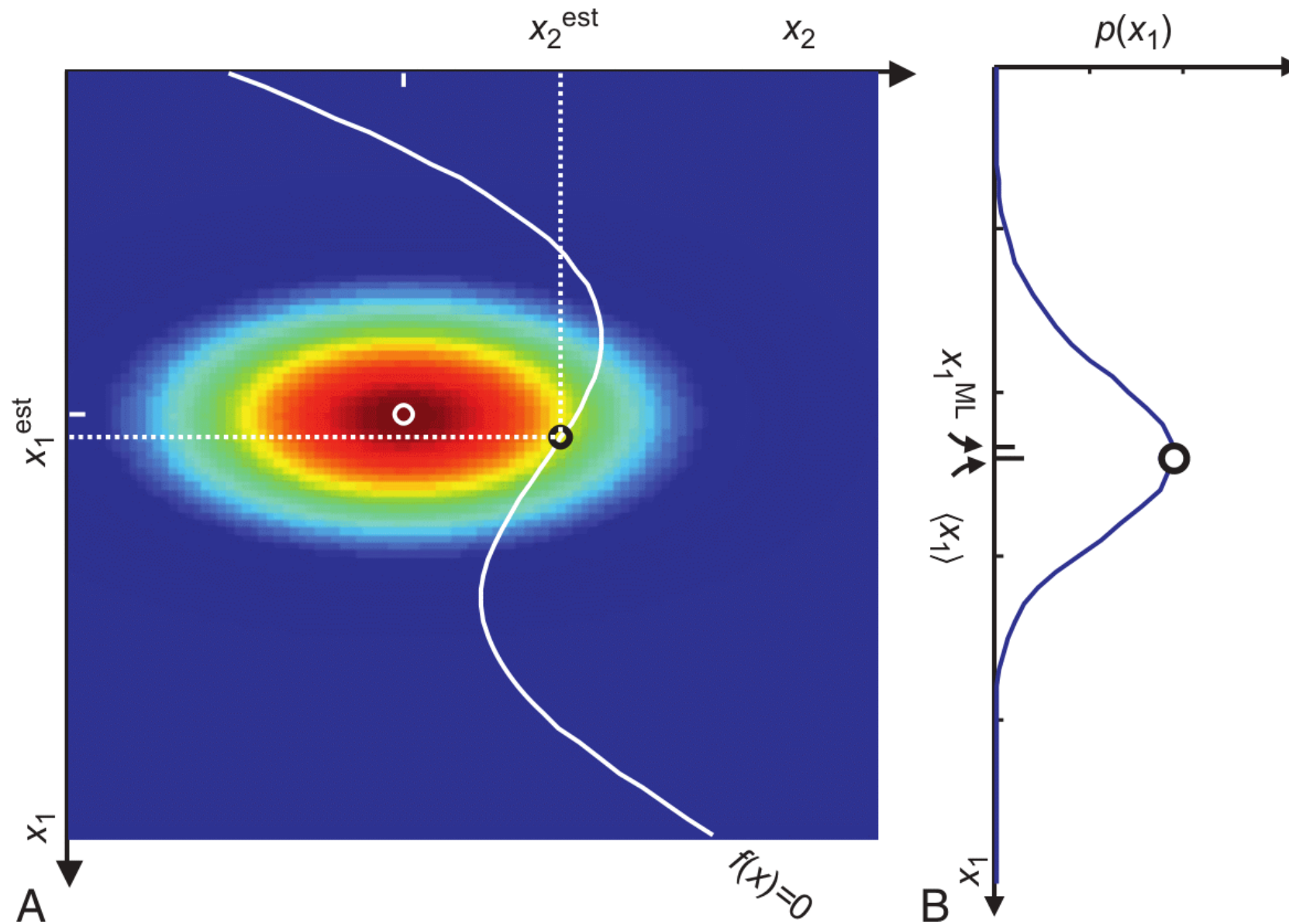


A: *a priori* pdf $p_a(\mathbf{m}, \mathbf{d})$, B: conditional pdf $p_g(\mathbf{m}, \mathbf{d})$, C: product $p_t(\mathbf{m}, \mathbf{d}) = p_a(\mathbf{m}, \mathbf{d})p_g(\mathbf{m}, \mathbf{d})$, white: theory

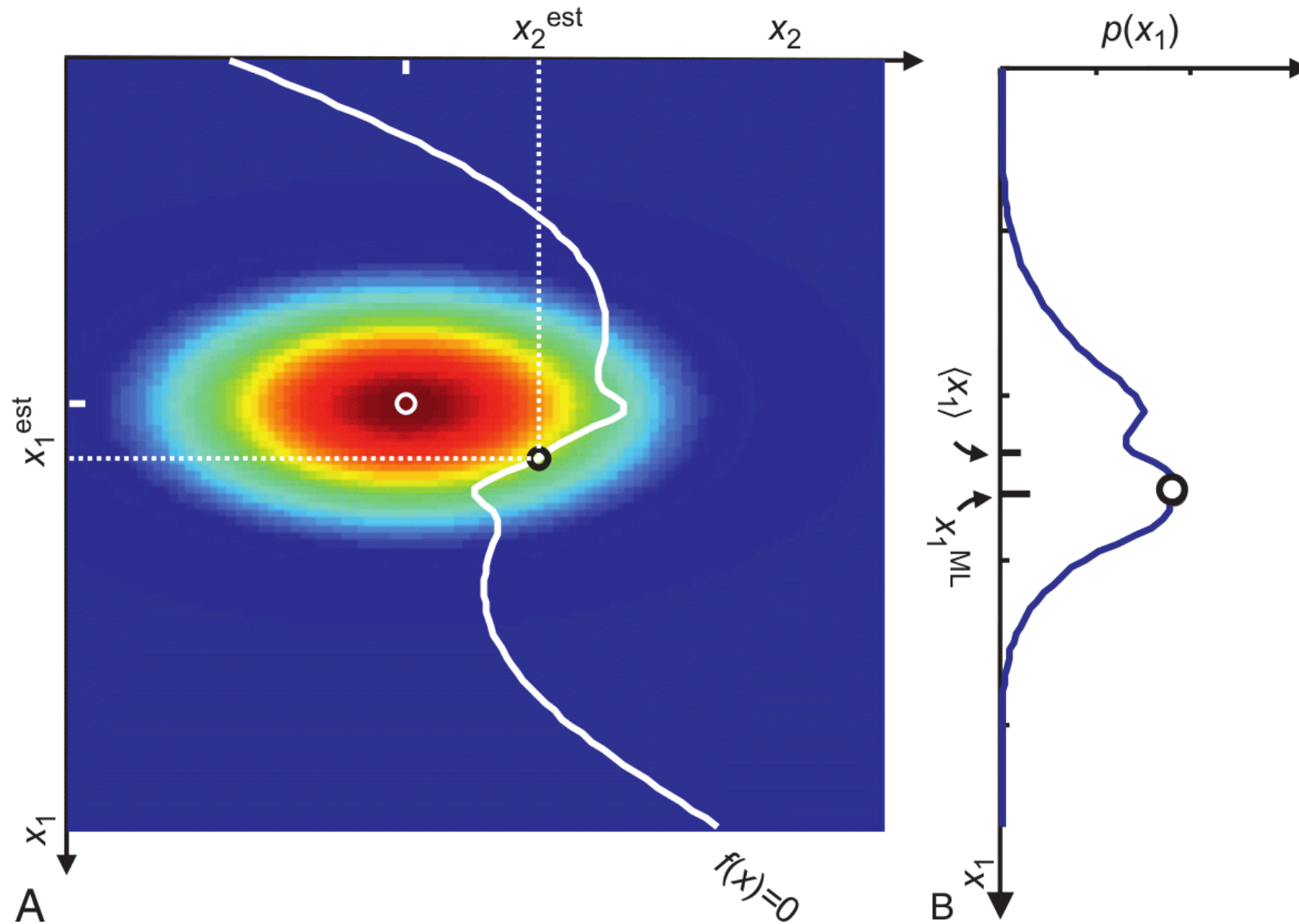
A priori and likelihood



Bayes view in nonlinear problems

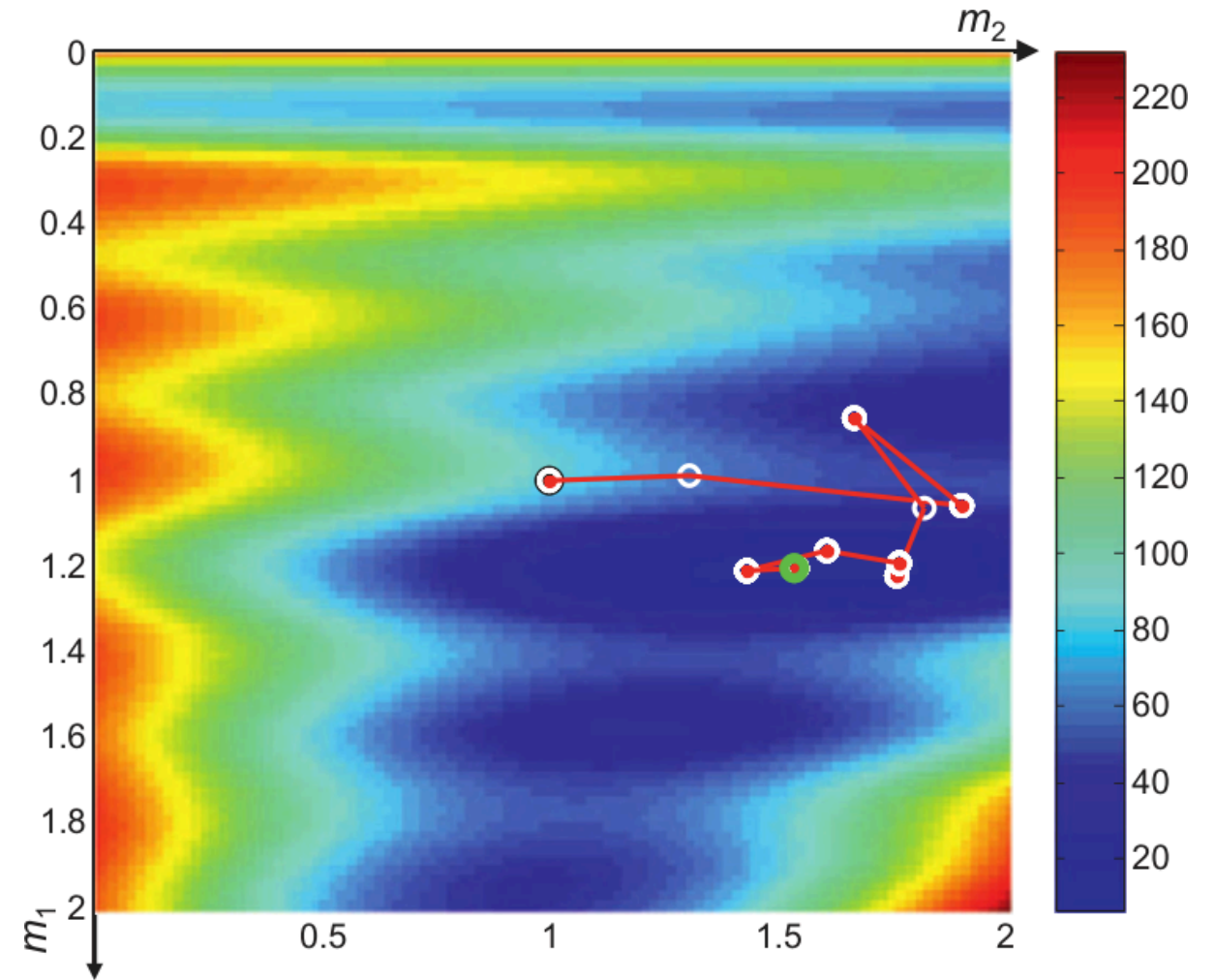


Highly nonlinear problems



Monte Carlo methods

- Monte Carlos search: randomly draw solutions from grid
- accept solution only if better than old
- Markow-Chain-Monte-Carlo
- Metropolis-Hastings (Metropolis et al., 1953; Hastings, 1970)

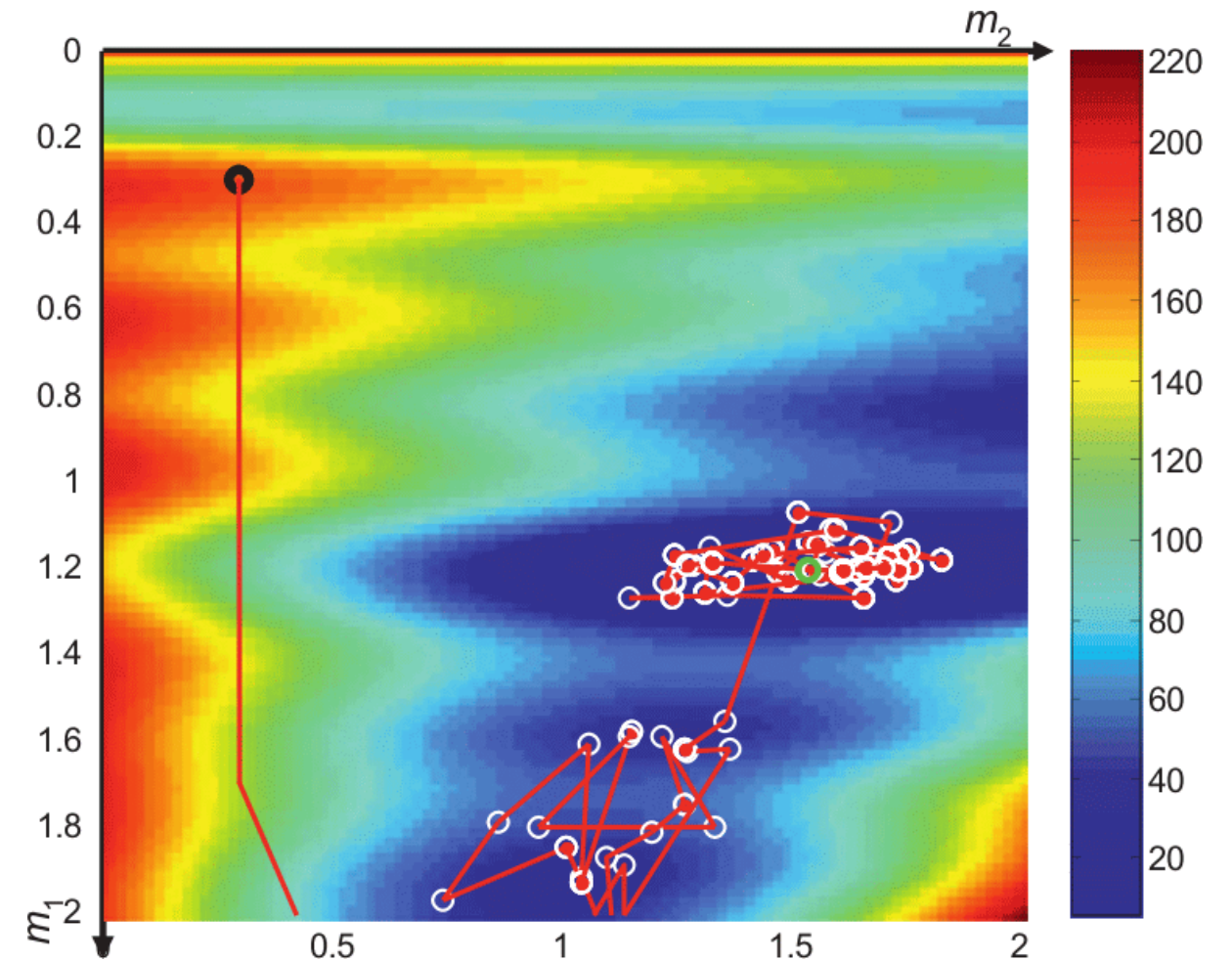


Monte Carlo method

Simulate Annealing

Test parameter

$$t = e^{-(\Phi(\mathbf{m}) - \Phi(\mathbf{m}^p))/T}$$



Simulated Annealing

Alternatives to grid search

i Monte Carlo search

draw random samples and accept them if the error is improved

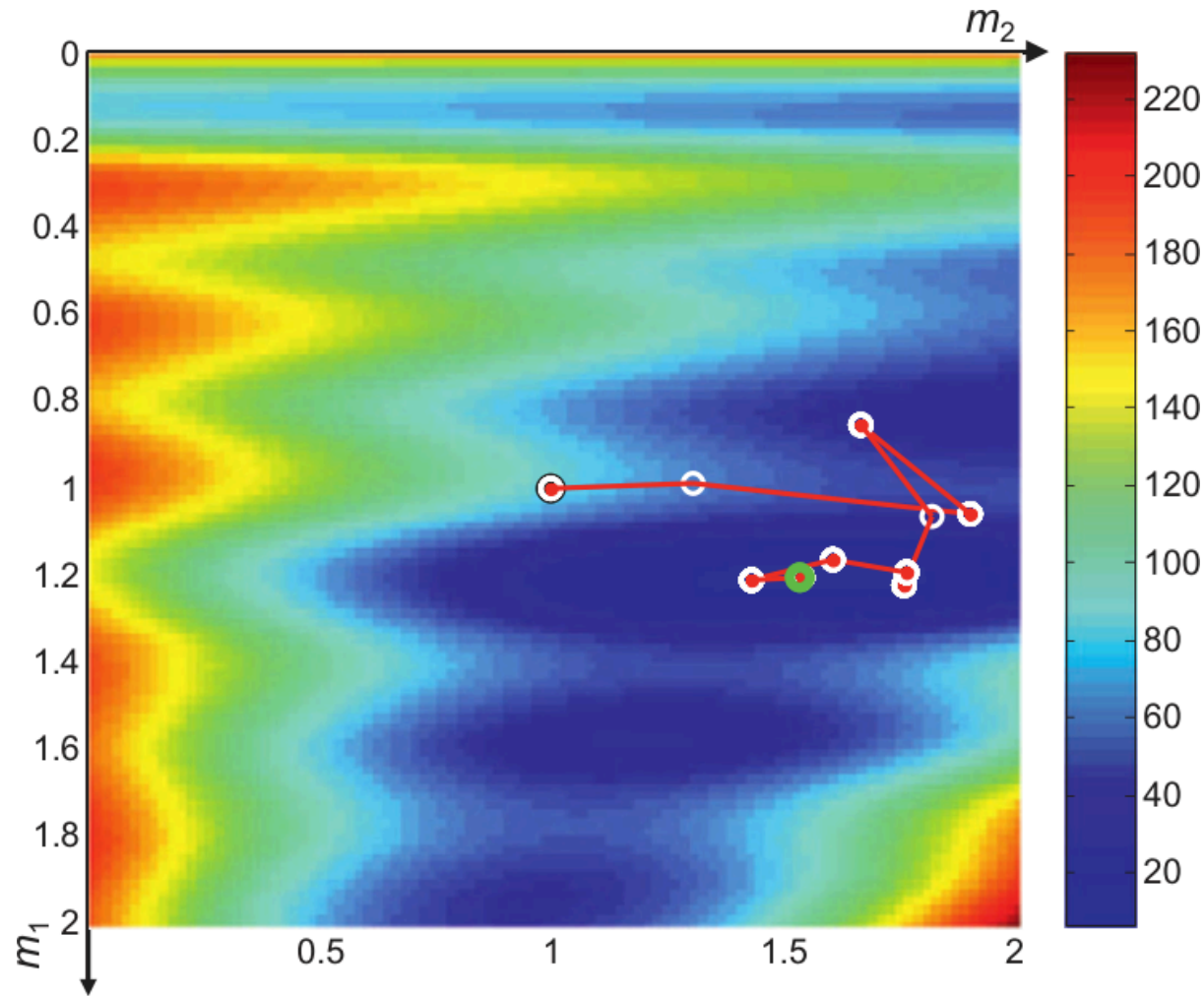
undirected search (Newtons method is directed)

i Simulated annealing

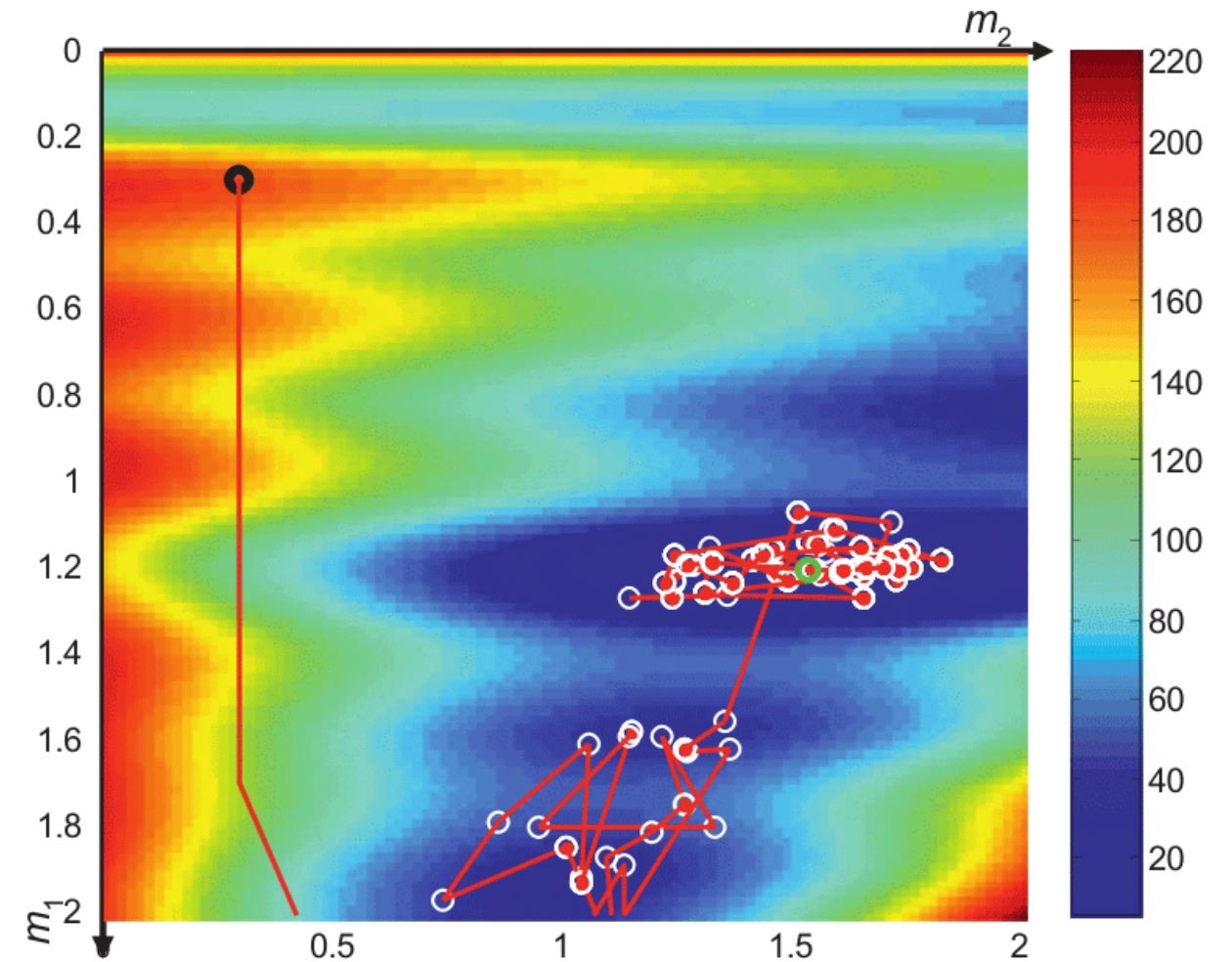
decrease temperature controlling particle movements:

high T : undirected, low T : search in vicinity of current model

Monte Carlo vs. Simulated Annealing



Monte Carlo method



Simulated Annealing